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Decoding the temporal dynamics of numerical semantics: Early cardinal activation and the critical role of ordinal processing in spatial-numerical associations

Zhiwei Zhang [a], Qingqing Chen [a], Chengtao Wang [a], Dongxue Zhao [a], Yun Pan [a,b,*]

[a] School of Psychology, Guizhou Normal University, Guiyang, China
[b] Guizhou Education University, Guiyang, China

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ABSTRACT

Numbers are fundamental to human culture and technological advancement, influencing everything from basic measurement to complex computations. Despite their ubiquity, the cognitive processes underpinning numerical understanding, especially the spatial cognition of numbers, remain inadequately explored. Understanding these processes is crucial as they underpin our ability to learn and interact with our environment. This study utilizes a dual choice go/nogo paradigm integrated with event-related potential (ERP) technique and multivariate pattern analysis (MVPA) to examine the semantics processing mechanisms within spatial-numerical associations (SNAs), particularly focusing on the dynamic temporal characteristics of cardinal and ordinal semantics. The behavioral results revealed that the Spatial-Numerical Association of Response Codes (SNARC) effect was observed only in Experiment 1, whereas the ordinal position effect emerged consistently across both experiments, suggesting that SNAs primarily originate from sequential information constructed in working memory, challenging the traditional view that SNAs primarily arise from the spatial distribution of numerical quantity. Traditional ERP analysis did not detect significant motor preparation linked to numerical semantics. However, MVPA results indicated that cardinal semantics are processed prior to ordinal semantics. This temporal dissociation was observed with the decoding accuracy of cardinal semantics being significantly higher than the chance at 112 ms post-stimulus. In contrast, ordinal semantics processing emerged later (around 180–200 milliseconds), and combined analysis with response-related ERP component demonstrated that SNAs show closer associations with ordinal semantics. This research enriches our understanding of numerical cognition and provides new perspectives for exploring the complexity and diversity of SNAs.

1. Introduction

Numbers play a crucial role in our daily lives, whether in measurement, ordering, or computation. Despite their pervasive use, we still lack a clear understanding of the mechanisms underlying numerical cognition. The symbolic representation of Arabic numerals and number words is unique to human culture, significantly contributing to technological advancement and shaping our modern society (Nieder & Dehaene, 2009). As early as the 17th century, Immanuel Kant linked spatial orientation to basic numerical computation, laying the groundwork for later theories of numerical cognition (Dehaene & Brannon, 2010). Galton (1880) explored visual representations of numbers in physical space, discovering significant variability among individuals in how they visualize numerical sequences. Dehaene et al. (1990, Dehaene et al., 1993) demonstrated that, in number magnitude classification or parity judgment tasks, participants responded faster with their left hands to small numbers and with their right hands to large numbers. This seminal finding, known as the Spatial-Numerical Association of Response Codes (SNARC) Effect, suggests the existence of Spatial-Numerical Associations (SNAs): small numbers are associated with the left side, and large numbers with the right. While the SNARC effect demonstrates consistency in populations with left-to-right reading habits, significant individual variability persists within these groups (Roth et al., 2024). The origins of this spatial representation remain debated. Some studies

* Correspondence author at: School of Psychology, Guizhou Normal University, No. 100, Huaxi University Town, Gui'an New District, Guiyang, Guizhou Province 550025, China.
E-mail address: panyun129@163.com (Y. Pan).

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suggest potential influences of cultural or reading direction (Dehaene et al., 1993; Shaki et al., 2009), while others report limited cultural or reading habit modulation (Hochman et al., 2025; Pitt & Casasanto, 2020). In recent years, the prevalence of the SNARC effect has been validated across various studies, including different experimental materials, response modalities, sensory channels, and participant groups (see reviews by Wood et al., 2008; Zhang et al., 2022).

Researchers have focused on the reasons for the SNARC effect, with the mental number line theory as the primary explanation (Dehaene, 2003). According to this theory, individuals mentally represent numerical magnitudes along a left-to-right spatial continuum, associating smaller numbers with the left side and larger numbers with the right. However, the theory encounters a confounding issue between two hypotheses: whether individuals implicitly map quantities into space or spatialize sequential relations between stimuli (Casasanto & Pitt, 2019). When extracting semantics information from symbolic numbers, there is no clear boundary between quantity and order. For example, the symbolic number “1” can be placed on the left side either because it represents a smaller quantity or because it is first in sequential order. It is unreasonable to attribute SNAs solely to the mapping of numerical quantities in space. Prpic et al. (2021) propose that both quantity and order are closely linked to the SNARC effect and highlight the necessity of considering stimulus type and task demands. This perspective is further supported by evidence from non-numeric domains: for instance, Prpic et al. (2016) demonstrated that musical note values can dissociate ordinal and magnitude processing through task demands, revealing distinct Order-Related and Magnitude-Related Mechanisms. Similar associations have been found for both numbers carrying quantity and order information, and for letters or months with only order information (Gevers et al., 2003; He et al., 2020). Therefore, distinguishing between the order and quantity of numbers is crucial to exploring SNAs.

Number cognition is essentially a process of activating cardinal and ordinal semantics by processing the quantity and order information of number (Sixtus et al., 2023). Cardinal semantics refer to the exact quantity of elements in a set (e.g., 1, 2, 3), while ordinal semantics refer to the position of elements in a sequence (e.g., first, second, third). Neuroimaging studies have shown different cognitive neural mechanisms for processing cardinal and ordinal semantics (Turconi et al., 2006). For instance, one frontal-injured patient was unable to understand the cardinal meaning of numbers and perform calculations but could correctly complete ordering tasks (Delazer & Butterworth, 1997). In contrast, a patient with Gerstmann syndrome could not process the order of numbers but had no impairment in quantity processing (Turconi & Seron, 2002). Recent studies have demonstrated that anodal tDCS over the left prefrontal cortex modulates the direction of SNARC-like effects for sequential stimuli (e.g., months) while showing no significant influence on numerical stimuli (Schroeder et al., 2017b). Subsequent research further established that this neuromodulatory effect is specifically observed in tasks requiring ordinal processing demands (Schroeder et al., 2017a). Functional magnetic resonance imaging (fMRI) studies revealed that the intraparietal sulcus region and frontoparietal network areas are primarily activated when processing cardinal semantics (Hubbard et al., 2005). In contrast, the anterior regions of the intraparietal sulcus are predominantly activated during ordinal semantics processing (Ischebeck et al., 2008; Kaufmann et al., 2009). These studies have shown that the intraparietal sulcus is a key node for number information processing (Nieder, 2025; Nieder & Dehaene, 2009). However, sequential processing of quantitative stimuli (e.g., numbers) and non-quantitative stimuli (e.g., letters) activates different brain regions (Zorzi et al., 2011). Although fMRI's high spatial resolution reveals cortical activity for cardinal and ordinal processing, the temporal overlap of these processes remains unclear (Jacob & Nieder, 2008). One study using event-related potential (ERP) technique compared the temporal dynamics of cardinal and ordinal processing phases (Turconi et al., 2004). Participants judged whether a number was greater or less than a reference number (quantity task) and whether a number was anterior or posterior to a reference number (ordering task). It found that cardinal processing preceded ordinal processing. However, the study struggled to fully separate the semantics processing of cardinal and ordinal numbers. Participants determining whether a number is anterior or posterior to another inevitably process quantity information first. Thus, it is unreasonable to assume that ordering task only activate ordinal processing; they also involve other cognitive processes, including quantity processing (Rubinsten & Sury, 2011). Individuals process semantics information about symbolic numbers without clear boundaries between cardinal and ordinal aspects. The ordering of objects often depends on the quantities they represent, with larger quantities placed later in the sequence (Lyons et al., 2016). Therefore, effectively separating cardinal and ordinal semantics is crucial for exploring the temporal dynamics of numerical semantics processing.

The dual choice go/nogo paradigm has been widely applied in psycholinguistics and combined with electrophysiological techniques, allows for the separation of the processing temporal dynamics of different mental activities. This is achieved by analyzing the Lateralized Readiness Potential (LRP) component, which is closely related to the selection and preparation of the contralateral response (Smulders & Miller, 2011). A negative deflection is observed in motor cortex electrodes when response preparation is completed, and the negative potential of the electrode on the contralateral side of the responding hand is greater than on the ipsilateral side (Fig. 1.A). In the dual choice go/nogo paradigm, participants made judgments about stimuli with two attributes: attribute A determined which hand to respond with, and attribute B determined whether to respond. If attribute A was processed before attribute B, response preparation based on attribute A occurred before determining whether to respond, allowing the LRP component to be detected on both go and nogo trials. As processing of attribute B is completed and the response hand is determined, the LRP of the nogo trial subsides while a persistent negative deflection of the LRP occurs during the go trial. Conversely, if attribute B is processed before attribute A, no motor preparation based on attribute A occurs until the response is determined, making it impossible to observe LRP on the nogo trial (Li et al., 2013). The LRP component has been widely used to study the processing temporal dynamics of various mental activities, including syntax and phonology (Turennout et al., 1998), gender and attraction (Carbon et al., 2018), style and content of art (Augustin et al., 2011), and gender and phonological extraction (Shantz & Tanner, 2020). Following the research methods outlined in earlier studies, evidence for the semantics processing temporal dynamics of numbers can be obtained by isolating the LRP component.

The present study adopts a dual choice go/nogo paradigm in which the parity judgment task is linked to cardinal semantics, while the memory sequence detection serves to probe ordinal semantics (Fig. 1.B). In the parity judgment task, although the decision about which key to press is directly based on classifying a number as even or odd, this process inherently requires assessing the numerical property of divisibility by two. This task was designed to engage cardinal semantics through automatic magnitude activation, as demonstrated by SNARC effects where numerical magnitude influences spatial responses even when irrelevant to the task (Cipora et al., 2021; Dehaene et al., 1993). Felisatti et al. (2024) observed that most studies employing parity judgments capture underlying number magnitude processing without explicit magnitude comparisons, thus underscoring the task's sensitivity to cardinal representations. Importantly, because resolving parity does not demand the construction or retrieval of an explicit sequence of numbers, it disengages the ordinal component of numerical representation (Mingolo et al., 2021). Conversely, the memory sequence detection explicitly engages ordinal processing through its working memory requirements—participants must dynamically maintain and retrieve transient numerical sequences, a paradigm shown to induce spatial coding based on ordinal positions rather than absolute magnitude (van Dijck & Fias, 2011). Numbers at the front of a memory sequence are consistently retrieved faster than those at the back, demonstrating that

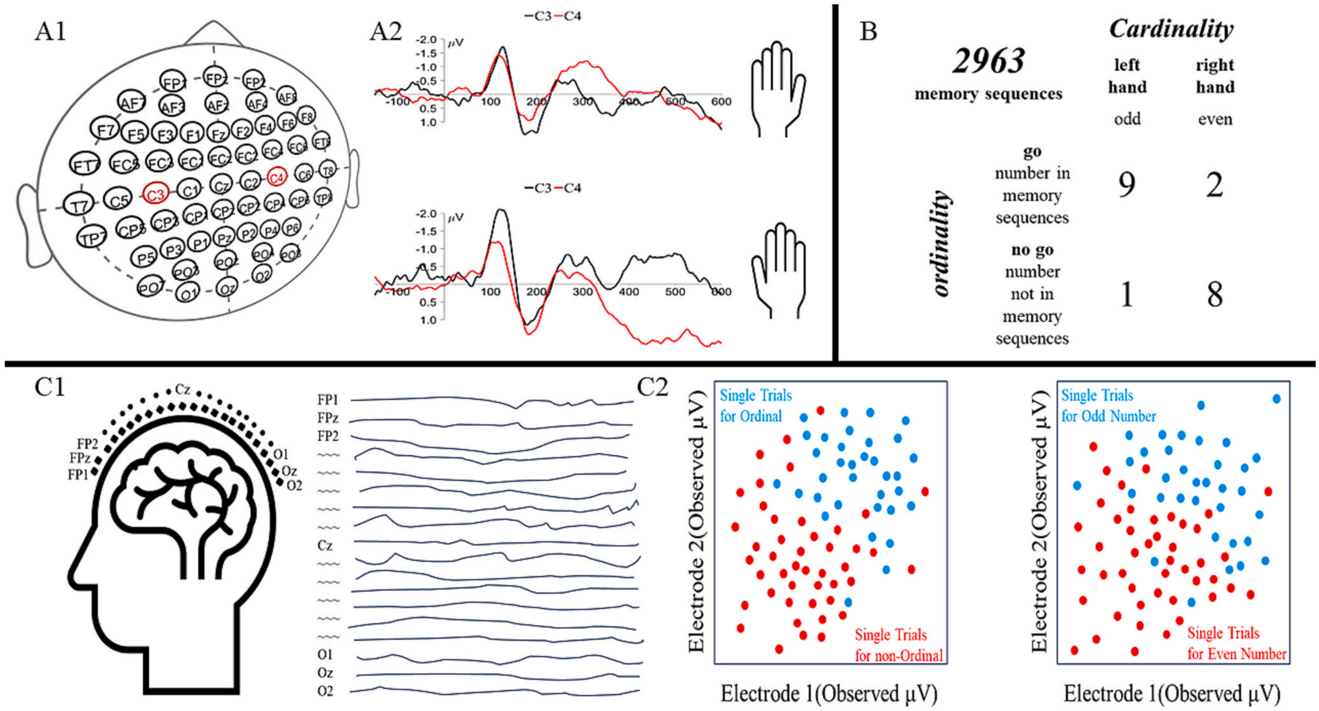

Fig. 1. Experimental Design and Methodological Comparison: Univariate Analysis Versus MVPA in EEG Research. (A) The A1 figure shows the primary electrodes for collecting scalp electrical signals from the participants, with C3 and C4 (marked in red) serving as the main electrodes for analyzing the LRP component. In the A2 figure, the electrodes on the contralateral side of the left and right hands induced greater negative deflection compared to those on the ipsilateral side during movement. B) Examples of the dual choice go/nogo paradigm are presented. The parity judgment task determines the response hand, while the memory sequence detection determines the go/nogo response. Participants first memorize the number sequence 2963. In the subsequent judgment task, if the stimulus number belongs to the memory sequence, participants respond with the left hand for odd numbers and the right hand for even numbers. If the number does not belong to the memory sequence, no response is made. C) The C1 figure illustrates the simultaneous recording of different voltage changes between each electrode. The C2 figure depicts the voltage distribution of two different types of stimuli on two electrodes at various time points. The essence of decoding lies in distinguishing between different types of stimuli as accurately as possible.

participants rely on the ordinal positions within the transiently constructed sequence to guide their responses. Furthermore, some studies suggest that accessing the ordinal positions of numbers depends on the spatial representation of the numerical sequence in the mind (Casasanto & Pitt, 2019; Koch et al., 2023). By clearly linking the parity task to automatic magnitude activation and the response selection in the go/nogo task to sequence-dependent positional retrieval, our design provides a robust theoretical framework that dissociates cardinal from ordinal semantics processing.

We hypothesized that subtle changes in scalp voltage would reflect temporal dynamics in semantics processing of numbers. However, traditional univariate ERP analysis has two limitations: first, it assumes uniform scalp voltage distributions among participants for a given constituent, despite significant individual variability in cortical folding patterns (Borrell, 2018). Second, electroencephalogram (EEG) studies often record data from over 30 scalp electrodes, yet analysis typically utilizes voltage values from only a few, leading to smaller effect sizes in statistical outcomes (Carrasco et al., 2024). The LRP component, derived from contralateral phase subtraction of C3 and C4 electrodes, eliminates motion-independent voltage deflections. However, the resulting smaller voltage differences can obscure significant experimental effects. To address these limitations, we performed multivariate pattern analysis (MVPA) on the voltage data recorded from all scalp electrodes (Fig. 1.C). MVPA has been widely adopted in neuroimaging studies and is a powerful tool for addressing complex theoretical questions in fMRI research (Fahrenfort et al., 2018; Zorzi et al., 2011). Recently, MVPA has also been widely utilized in EEG research. Researchers have found that by decoding voltages from whole-brain recording electrodes, they can successfully identify differences in EEG activity related to facial identity and emotion, as well as in motor perception, semantics processing, and working memory (Bae, 2021; Bae & Luck, 2019; He et al., 2024; Smith & Smith, 2019; Trammel et al., 2023; Yu et al., 2023). Therefore, employing machine learning to decode scalp potentials from multiple electrodes can reveal finer-grained temporal differences in number semantics processing compared to traditional univariate ERP analyses.

Building on the experimental paradigm established by van Dijck and Fias (2011), this study investigates whether SNAs primarily depend on automatic spatial representations of numerical quantity or dynamically constructed working memory sequences. By integrating ERP technique and MVPA across two experiments with divergent task requirements, we systematically examine the temporal dynamics of cardinal and ordinal semantics processing. Specifically, Experiment 1 employs a design where ordinal semantics (sequence position) governs response initiation (go/no-go), while cardinal semantics (automatic quantity processing in parity judgment) determines manual response selection (left/right keypresses). This relationship is reversed in Experiment 2, where cardinal semantics controls response initiation and ordinal semantics directs response selection. Such counterbalanced task configurations mitigate potential confounds in interpreting semantics processing dynamics induced by task-specific demands. We hypothesize that behavioral results will demonstrate participants' reliance on ordinal information extracted from working memory representations, consistent with prior evidence suggesting the dominance of ordinal semantics in SNAs (Ginsburg et al., 2014; van Dijck et al., 2013; van Dijck & Fias, 2011). Furthermore, given that parity judgments inherently engage cardinal semantics processing through automatic magnitude activation (Cipora et al., 2021), and SNARC effects are temporally coupled with response selection stages (Keus et al., 2005; Pan et al., 2022). Univariate ERP analyses and MVPA decoding should reveal: (1) earlier neural signatures of cardinal processing compared to ordinal semantics, and (2) stronger associations between certain semantics and response selection phases. This research aims to elucidate both the temporal dynamics of numerical semantics processing and the relative contributions of cardinal versus ordinal information in shaping SNAs.

2. Experiment 1

The purpose of this experiment was to investigate the temporal dynamics of cardinal and ordinal semantics processing in numerical cognition, and to determine whether numerical spatial representations are driven by quantity or working memory sequences.

2.1. Methods

2.1.1. Participants

In accordance with the methodology described by Pinto et al. (2018), we employed the G*Power 3.1 software for statistical power analysis to ascertain the necessary sample size for our study. For the two within-subject factors repeated-measures ANOVAs, we specified a power of 0.85, an alpha level of 0.05 (two-tailed), and an effect size of 0.25. Although the analysis indicated a requirement of 26 participants, we opted to include 31 participants in the experiment to ensure robustness. Thirty-one participants from Guizhou Normal University participated in this experiment, including 16 female students with a mean age of 21.42 years ($SD = 2.25$). All participants were right-handed, reported normal or corrected vision, had no history of mental illness, and were in good health. They signed an informed consent form prior to the experiment and were compensated at the end. This study was approved by the Committee for the Protection of Human Subjects, School of Psychology, Guizhou Normal University (Review number: GZNUPSY.N.202406 E [0018]).

2.1.2. Apparatus and stimuli

The experimental program was developed using MATLAB and the Psychotoolbox toolkit. A 24-in. monitor with a resolution of 1920 × 1080 and a refresh rate of 60 Hz displayed the stimuli. Participants' eyes were positioned 75 cm from the monitor, ensuring a viewing angle of 1.5° for both numerical stimuli and the gaze point. Stimuli comprised numbers 1–9 (excluding 5), presented against a black background. All experimental sessions took place in a well-lit room.

2.1.3. Design

The experiment employed a within-subject design featuring two key manipulations: (1) a 2 (magnitude: small = 1–4 vs. large = 6–9) × 2 (response hand: left vs. right) factorial design to examine whether the spatial representation of numbers is based on quantity, and (2) a 2 (ordinal position: initial = first two numbers vs. later = last two numbers) × 2 (response hand: left vs. right) factorial design to assess whether the spatial representation of numbers is based on the order in working memory. This representation of numerical space based on working memory sequence is called ordinal position effect.

2.1.4. Procedure

The experiment comprised 48 blocks, each containing 16 parity judgment trials. Within each block, eight unique numbers were presented twice in randomized order. Each block was divided into four phases: encoding, maintain, classification, and verification, see Fig. 2. In the encoding phase, four non-repeating numbers were presented sequentially in the center of the screen. To ensure participants could retain the number sequence, the next number was presented only after the space bar was pressed. Numbers were presented in strict odd-even alternation, with consecutive numbers never sharing the same parity. In the maintain phase, a blank screen was presented for 2500 ms, allowing participants to rehearse the number sequence. In the classification phase, each of the eight numbers appeared twice at random. Participants judged whether numbers from the memorized sequence were odd or even. For half of the experiment, they pressed the left key for odd numbers and right key for even numbers. For the other half, this left-right assignment was reversed. If the number was not in the memory sequence, no response was required. Parity judgment required both speed and accuracy, with a reaction time (RT) limit of 1500 ms. During the verification phase, participants had to choose the correct memorized sequence from three similar-looking number sequences. If the wrong sequence was selected, the entire block was repeated until the correct one was chosen. Prior to the main experiment, participants performed two practice blocks and were required to demonstrate at least 90 % accuracy before advancing to formal testing.

2.1.5. EEG recording and data preprocessing

This experiment used the Curry 8 EEG system with a 64-channel electrode cap arranged in the 10–20 standard system to record EEG data. The Cz electrode on the top of the head served as the online reference, and the electrode between the Fz and FPz electrodes was grounded. Bipolar recordings of horizontal and vertical electrooculograms (EOGs) were made, with electrodes for horizontal EOG placed 1 cm from the left and right eyes and electrodes for vertical EOG

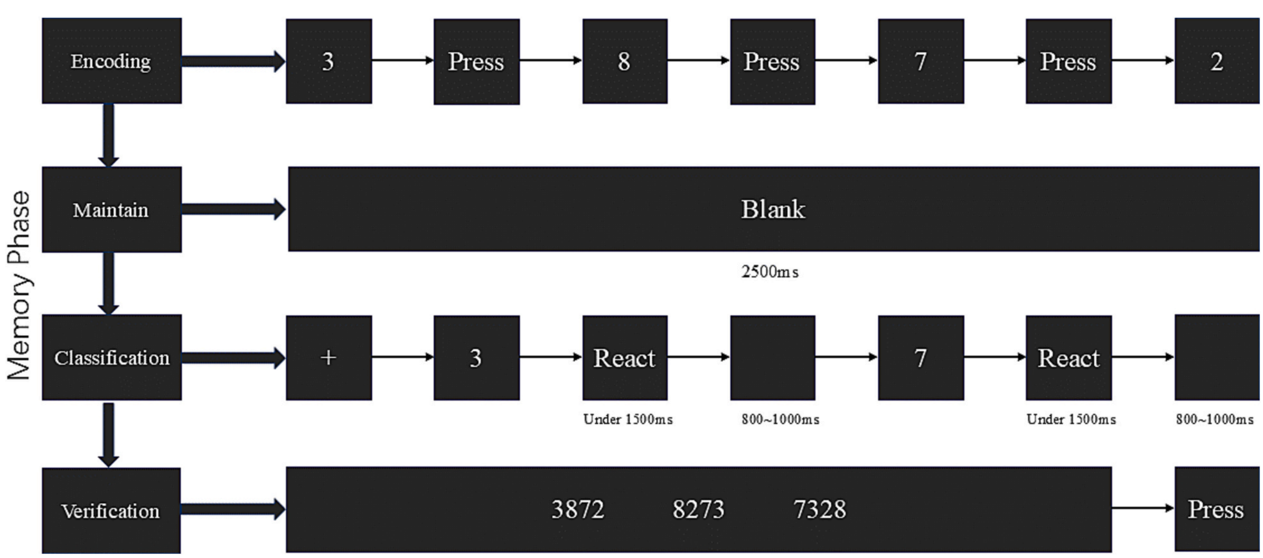

Fig. 2. Time Flow of Trial Events in Experiment 1. *Note.* Details can be found in the description of the procedure section.

positioned 1 cm vertically above and below the left eye. The sampling rate was 1000 Hz with a bandpass filter of 0.05–100 Hz, and all scalp impedances were below 10 kΩ. Offline analysis was conducted using the EEGLAB and ERPLAB toolboxes in MATLAB (Delorme & Makeig, 2004; Lopez-Calderon & Luck, 2014). Applied an offline bandpass filter was set to 0.1–30 Hz, and the data were re-referenced to the average of all electrodes. Independent component analysis was employed to correct for ocular artifacts. EEG segments of 1000 ms were extracted, with the 200 ms preceding the stimulus presentation serving as the baseline and the 800 ms following the stimulus presentation as the analysis epoch. Trials with amplitudes exceeding ±75 μV were excluded.

2.1.6. Univariate ERP analysis

To investigate the temporal dynamics of processing cardinal and ordinal semantics, experimental trials were divided into go-trials and nogo-trials. Go-trials with response timeouts or incorrect responses were excluded, as were nogo-trials with any responses. The LRP was recorded at the C3 and C4 electrodes and isolated using a double subtraction method:

$$\text{LRP} = (\text{C3} - \text{C4})\ \text{right hand}/2 - (\text{C3} - \text{C4})\ \text{left hand}/2$$

The average voltage of C3-C4 was calculated separately for right-hand and left-hand trials. The LRP was then derived by subtracting the left-hand average from the right-hand average. This calculation eliminated voltage asymmetries unrelated to response preparation. To determine the significance of the LRP component, a stepwise *t*-test method was used (Li et al., 2013). Starting from stimulus onset, *t*-tests with a 10 ms step and a 50 ms analysis window were conducted to calculate the mean voltage for each time window (e.g., 150–200 ms, 160–210 ms). One-tailed *t*-tests with a 95 % confidence interval were performed for each window to test whether the mean voltage within the window significantly differed from the baseline. If five consecutive windows showed significant differences, the presence of an LRP component was indicated. The onset latency of the LRP component was determined by the first significant time window.

2.1.7. MVPA

The MVPA-light toolbox was used for MVPA analysis (Treder, 2020). To improve computational efficiency, the sampling rate was reduced to 250 Hz, and horizontal and vertical EOG electrodes were excluded. Other preprocessing steps were consistent with those used in the univariate analysis. Voltage values at 250 time points from −200 ms to 800 ms were used for independent decoding. Each correct go or nogo trial was classified based on cardinal and ordinal semantics (e.g., for the memorized sequence 9278, the 7 was classified as ordinal and odd, the 8 as ordinal and even, the 3 as non-ordinal and odd, and the 6 as non-ordinal and even). Single-trial ERP data are noisy; thus, averaging trials before MVPA improves statistical power (Carrasco et al., 2024; Trammel et al., 2023). We averaged 20 trials for each cardinal and ordinal semantics condition, further dividing each condition into two categories (ordinal semantics: ordinal vs. non-ordinal; cardinal semantics: odd vs. even). A Logistic Regression classifier was used to decode each participant's EEG data, with 10-fold cross-validation repeated 100 times. Classification performance was expressed as accuracy, with a chance probability of 50 %.

Temporal generalization was explored to determine whether numerical semantics decoding was consistent across time points, the classifier trained at each time point was tested at all other time points to investigate the generalization performance of numerical semantics decoding over time. We also examined the contribution of scalp electrodes to numerical semantics decoding. To assess whether decoding results were significantly above chance, we conducted a non-parametric cluster-based permutation test, correcting *t*-test results for multiple comparisons using a reference distribution from 1000 iterations, with a Wilcoxon test threshold of $p = 0.05$. All MVPA parameters were kept consistent.

2.2. Results

2.2.1. Behavioral data

All data exceeding 3 standard deviations from the mean RT, incorrect responses, and non-responses were excluded, accounting for 2.97 % of the data. No participants with an error rate over 10 % were excluded from the analysis. Behavioral data were analyzed using SPSS 27.

To investigate the presence of ordinal position effect, a repeated-measures ANOVA on mean RT with ordinal position × response hand was conducted. The analysis showed a significant main effect of ordinal position, $F(1,30) = 26.94$, $p < 0.001$, $\eta_p^2 = 0.47$. Responses to the initial numbers in the sequence (M = 771.78 ms, SE = 15.06 ms) were significantly faster than responses to the later numbers (M = 795.79 ms, SE = 16.02 ms). The main effect of response hand was not significant, $p = 0.717$. However, there was a significant interaction between ordinal position and response hand, $F(1, 30) = 14.50$, $p = 0.001$, $\eta_p^2 = 0.33$. Post hoc simple effects analysis shows that responses to the initial numbers in the sequence were faster with the left hand (M = 763.57 ms, SE = 16.21 ms) than the right hand (M = 779.99 ms, SE = 14.52 ms), $p = 0.014$. Responses to the later numbers in the sequence were faster with the right hand (M = 789.57 ms, SE = 15.29 ms) than the left hand (M = 802.01 ms, SE = 17.42 ms), $p = 0.082$, indicating the presence of ordinal position effect.

To investigate the presence of a SNARC effect, a repeated-measures ANOVA on mean RT with number magnitude × response hand was conducted. The main effects of number magnitude and response hand were not significant, $ps > 0.727$. However, the interaction between number magnitude and response hand was significant, $F(1, 30) = 5.59$, $p = 0.025$, $\eta_p^2 = 0.16$. Post hoc simple effects analysis shows that responses to small numbers were faster with the left hand (M = 778.22 ms, SE = 17.27 ms) than the right hand (M = 789.89 ms, SE = 14.54 ms), $p = 0.126$ and responses to large numbers were faster with the right hand (M = 779.75 ms, SE = 15.02 ms) than the left hand (M = 787.57 ms, SE = 16.58 ms), $p = 0.218$, indicating the presence of SNARC effect (see Fig. 3).

2.2.2. ERP results

A stepwise *t*-test was used to determine whether the LRP was present in both go and nogo trials. In go trials, the amplitude of the LRP was significantly different from zero at 220 ms post-stimulus, $t(30) = -2.15$, $p = 0.040$, 95 % CI [−0.230, −0.006]. However, in nogo trials, the amplitude of the LRP was not significantly different from zero, $ps > 0.010$, indicating no significant LRP component in nogo trials. The results of Experiment 1 suggest that the recognition of parity (response hand) was not completed before participants processed ordinal (go/nogo).

A notable feature of the LRP component is its relatively small effect size (1–4 μV), typically requiring 50–100 trials per reactive hand for sufficient superposition (Smulders & Miller, 2011). Although our experimental trials largely met these requirements, the LRP component, being based on EEG data from multiple trials, required contralateral phase subtraction to minimize the influence of action-independent components (Boudewyn et al., 2018). A significant limitation of this analysis was the use of only two electrodes (C3 and C4), which did not account for voltage differences at other scalp sites during the processing of ordinal and cardinal semantics. Therefore, we adopted the MVPA approach to decode and analyze whole-brain electrode activity during the processing of ordinal and cardinal semantics.

2.2.3. MVPA results

Decoding across time points demonstrated that the probability of decoding the ordinal semantics was significantly higher than the chance at 180 ms after stimulus presentation, with an average accuracy of 0.867 ($ACC_{min} = 0.533$, $ACC_{max} = 0.950$). Cross-electrode decoding demonstrated that the electrodes with the highest contribution to decoding ordinal semantics was located at the combination of the left occipital

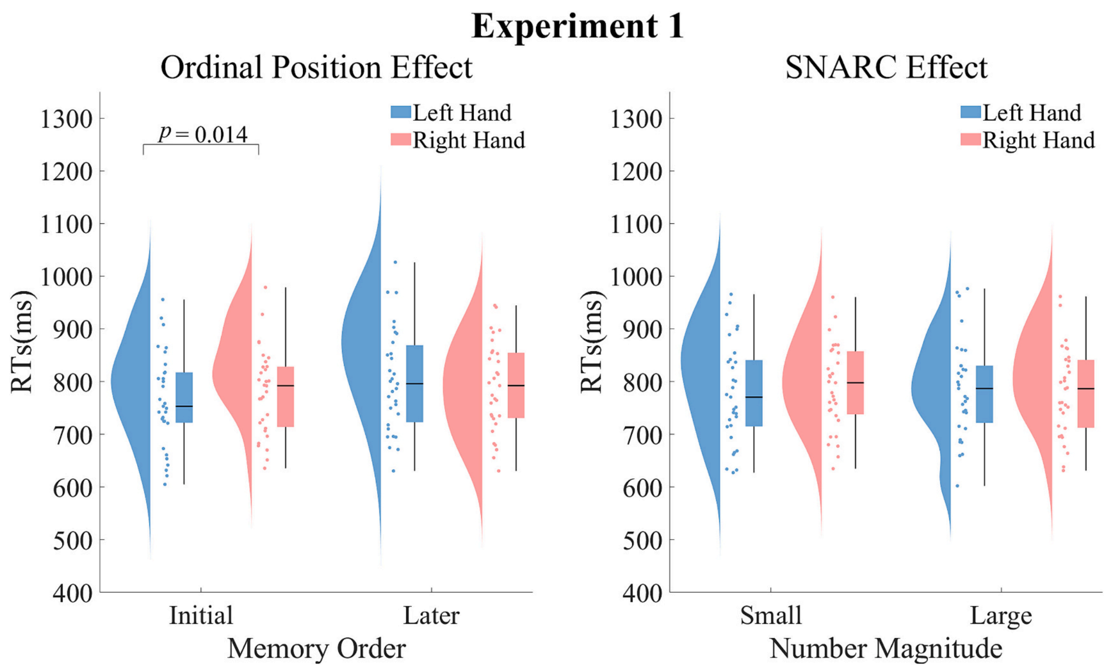

Fig. 3. Difference in RTs in Experiment 1. *Note.* The comparison of RTs between initial and later memory sequences in ordinal position effect and between small and large numbers in SNARC effect from Experiment 1.

and occipital parietal lobes. The results of temporal generalization demonstrated that approximately 31.49 % of the ACCs for ordinal semantics were significant, with an average accuracy of 0.735 (ACC_{min} = 0.510, ACC_{max} = 0.951). After 200 ms post-stimulus presentation, significant off-diagonal decoding was observed, especially after 300 ms, where decoding could be generalized to any time point thereafter.

Significant above-chance decoding of cardinal semantics occurred primarily in two temporal clusters across time points: 112 ms–156 ms and 196 ms–468 ms after stimulus presentation, with an average accuracy of 0.560 (ACC_{min} = 0.528, ACC_{max} = 0.595). Cross-electrode decoding demonstrated that the electrodes contributing most to the probability of decoding cardinal semantics was located in the left occipital lobe. The results of temporal generalization demonstrated that approximately 8.55 % of ACCs for cardinal semantics were significant, with an average accuracy of 0.543 (ACC_{min} = 0.515, ACC_{max} = 0.605). Significant diagonal decoding was demonstrated at 164 ms post-stimulus presentation, suggesting a time point of significant decoding without generalization performance at other time points. Fig. 4 presents

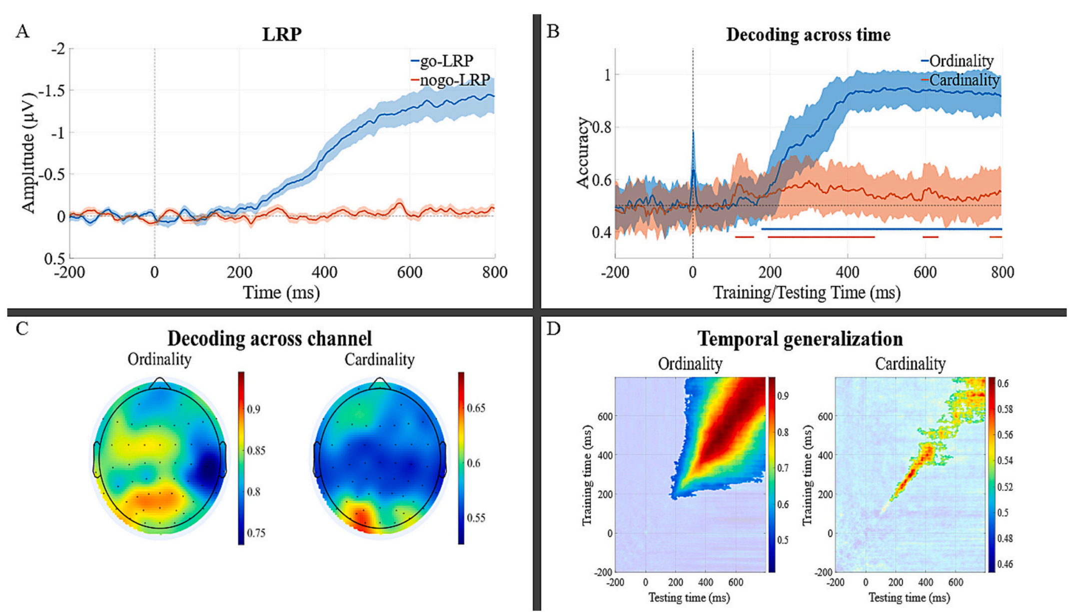

Fig. 4. Results of Univariate Analysis and MVPA of EEG in Experiment 1. Note: (A) Grand average LRPs for go and nogo trials. Shaded areas represent the standard error of the mean (SEM). Significant LRPs were observed in go trials, but not in nogo trials. B) Time-resolved decoding of numerical semantics information. Shaded areas indicate the SEM of decoding accuracy. The horizontal and colored solid lines at the bottom mark the time windows during which decoding accuracy for distinct semantics categories significantly differed from chance level (p < 0.05, corrected). C) Cross-electrode decoding of numerical semantics information. D) Temporal generalization results for numerical semantic decoding. Classifiers were trained at time points along the y-axis and tested at time points along the x-axis. Highlighted regions represent time windows where decoding accuracy was significantly higher than chance level (p < 0.05, corrected).

the results of the univariate analysis and MVPA of EEG.

2.3. Discussion

The behavioral results of Experiment 1 demonstrate that the spatial representation of numerical magnitude is relatively attenuated when a temporal numerical sequence is actively constructed within the task. Specifically, although a SNARC effect was observed ($\eta_p^2 = 0.16$), the ordinal position effect was considerably larger ($\eta_p^2 = 0.33$). Importantly, we measured the temporal dynamics of semantics processing of numbers using electrophysiological techniques. Traditional univariate analyses, which probe the semantics information of numbers using LRP, have not yielded reliable conclusions. However, the MVPA results from decoding EEG data revealed that the processing of cardinal semantics precedes ordinal semantics. A significant consideration is the assumption in Experiment 1 that the determination of whether to respond precedes the choice of which hand to use for the response. The disadvantage of this design is that if reactions are performed such that the reacting hand is determined before the judgment of whether to react is made, it is unreliable to conclude that the processing of cardinal semantics takes precedence over ordinal semantics. Therefore, further exploration is needed to determine whether the hypothesized reaction order affects the experimental results.

3. Experiment 2

The results of Experiment 1 showed the temporal advantage of cardinal semantics over ordinal semantics. To verify this finding, we conducted Experiment 2. In this experiment, we switched the roles of ordinal and parity judgments. The parity of the number determined whether a go/nogo response was required, and the memory sequence detection determined which hand was used for the response. This manipulation allowed us to assess whether the results observed in Experiment 1 were due to task demands.

3.1. Methods

3.1.1. Participants

Thirty participants from Guizhou Normal University participated in this experiment, including 17 female students with a mean age of 20.9 years ($SD = 2.01$). Other contents are consistent with Experiment 1.

3.1.2. Apparatus, stimuli, and design

Consistent with Experiment 1.

3.1.3. Procedure

The experiment comprised 48 blocks, each containing 16 parity judgment trials, eight numbers was presented twice at random. The encoding, maintain, and verification phases of the experiment were consistent with Experiment 1. In the classification phase, participants were first asked to make parity judgments on the presented numbers, responding only to odd numbers in the first 24 blocks. They responded with the left hand if the presented odd number belonged to the memorized sequence, and with the right hand if it did not. No response was required if the presented number was even. In the last 24 blocks, participants reacted only to even numbers, using the right hand if the presented even number belonged to the memorized sequence, and the left hand if it did not. No response was required if the presented number was odd. The judgment required both speed and accuracy, with a RT limit of 1500 ms.

3.1.4. Univariate ERP analysis and MVPA

Consistent with Experiment 1.

3.2. Results

3.2.1. Behavioral results

All data exceeding 3 standard deviations from the mean RT, incorrect responses, and non-responses were excluded, accounting for 2.40 % of the data. No participants with an error rate over 10 % were excluded from the analysis. Behavioral data were analyzed using SPSS 27.

To investigate the presence of ordinal position effect, a repeated-measures ANOVA on mean RT with ordinal position × response hand was conducted. The analysis showed a nonsignificant main effect of ordinal position, $p = 0.160$. The main effect of response hand was significant, $F(1, 29) = 8.53$, $p = 0.007$, $\eta_p^2 = 0.23$, with slower RTs for the left-hand ($M = 715.13$ ms, $SE = 18.79$ ms) than for the right-hand ($M = 687.30$ ms, $SE = 17.47$ ms). There was a significant interaction between ordinal position and response hand, $F(1, 29) = 22.44$, $p < 0.001$, $\eta_p^2 = 0.44$. Although the interaction between ordinal position and response hand was significant, post hoc analysis revealed no significant difference in RTs between hands for initial numbers, left hand ($M = 697.98$ ms, $SE = 17.53$ ms) slower than the right hand ($M = 694.58$ ms, $SE = 16.79$ ms), $p = 0.745$. The ordinal position effect was exclusively driven by later numbers, with significantly faster right-hand responses ($M = 680.03$ ms, $SE = 19.11$ ms) compared to left-hand ($M = 732.29$ ms, $SE = 20.86$ ms), $p < 0.001$, indicating the presence of ordinal position effect.

To investigate the presence of a SNARC effect, a repeated-measures ANOVA on mean RT with number magnitude × response hand was conducted. The analysis showed a significant main effect of number magnitude, $F(1, 29) = 7.77$, $p = 0.009$, $\eta_p^2 = 0.21$. RTs were longer for smaller numbers ($M = 749.12$ ms, $SE = 16.80$ ms) compared to larger numbers ($M = 738.45$ ms, $SE = 17.23$ ms). The remaining main effect and interaction were not significant, with *p*-values greater than 0.080, indicating no SNARC effect, see Fig. 5.

3.2.2. ERP results

Stepwise *t*-tests were used to explore whether LRP was present in both go and nogo trials. In go trials, the amplitude of the LRP at 300 ms post-stimulus was significantly different from zero, $t(29) = -2.13$, $p = 0.042$, 95 % CI [−0.331, −0.006]. However, in nogo trials, the amplitude of the LRP was not significantly different from zero, $ps > 0.05$, suggesting no significant LRP component in nogo trials. The results of Experiment 2 suggest that the processed ordinal (response hand) was not completed before participants recognition of parity (go/nogo). Considering the limitations of traditional univariate analyses discussed in Experiment 1, we also conducted an MVPA analysis to further explore these findings.

3.2.3. MVPA results

Decoding across time points demonstrated that the probability of decoding the ordinal semantics was significantly higher than the chance at 200 ms after stimulus presentation, with an average accuracy of 0.621 ($ACC_{min} = 0.532$, $ACC_{max} = 0.692$). Cross-electrode decoding showed that the electrodes with the highest contribution to decoding ordinal semantics was located at the left temporal and occipital parietal lobes. The results of temporal generalization demonstrated that ordinal semantics were significant for approximately 16.98 % of the ACCs, with an average accuracy of 0.584 ($ACC_{min} = 0.516$, $ACC_{max} = 0.702$). Significant off-diagonal decoding occurred at 200 ms after stimulus presentation, especially after 300 ms, where decoding could be generalized over a range of 300 ms from the test time point.

Significant above-chance decoding of cardinal semantics occurred primarily in two temporal clusters across time points: 112 ms–176 ms and 184 ms after stimulus presentation, with an average accuracy of 0.622 ($ACC_{min} = 0.546$, $ACC_{max} = 0.671$). Cross-electrode decoding showed that the electrodes contributing the highest probability to decoding cardinal semantics were located in the bilateral occipital lobes. The results of temporal generalization demonstrated that approximately 21.70 % of ACCs for cardinal semantics were significant, with an

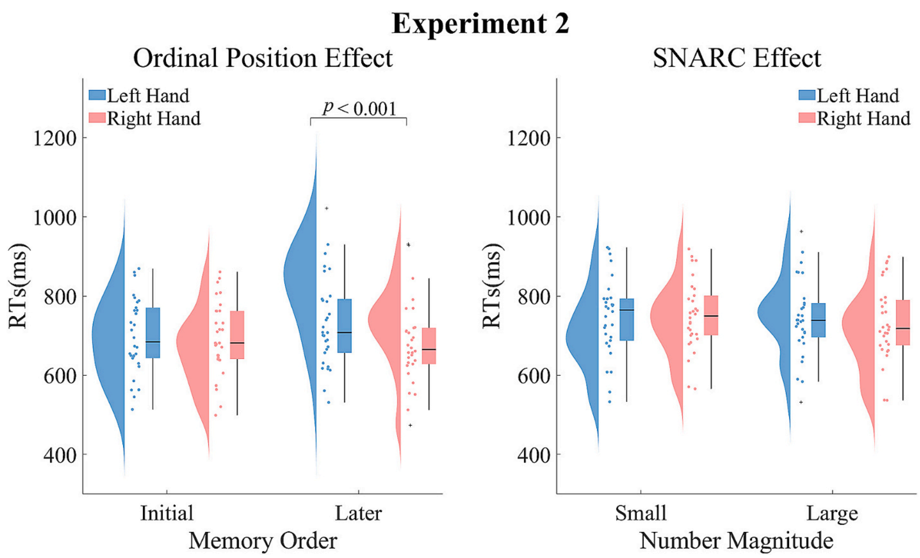

Fig. 5. Difference in RTs in Experiment 2. *Note.* The comparison of RTs between initial and later memory sequences in ordinal position effect and between small and large numbers in SNARC effect from Experiment 2.

average accuracy of 0.575 (ACC_{min} = 0.508, ACC_{max} = 0.670). The range of 100 ms to 300 ms after stimulus presentation exhibited significant diagonal decoding, suggesting specific timepoints had no generalization performance at other times. However, significant non-diagonal decoding patterns were also observed after 300 ms. Fig. 6 presents the results of the univariate analysis and MVPA of EEG.

3.3. Discussion

The behavioral results of Experiment 2 demonstrate an ordinal position effect but an absence of the SNARC effect, suggesting that SNAs arises from constructed ordinal information rather than numerical quantity representation. Importantly, we measured the temporal dynamics of semantics processing of numbers using electrophysiological techniques. Univariate ERP analysis showed that ordinal semantics did not induce early activation in the motor preparation stage. However, by decoding the temporal dynamics of numerical semantics processing, we observed that the processing of cardinal semantics precedes that of ordinal semantics. The following section will delve deeper into this finding.

4. General discussion

This study used a dual choice go/nogo paradigm to explore the representation of SNAs and the temporal dynamics of cardinal and ordinal semantics. The behavioral results revealed that the SNARC effect

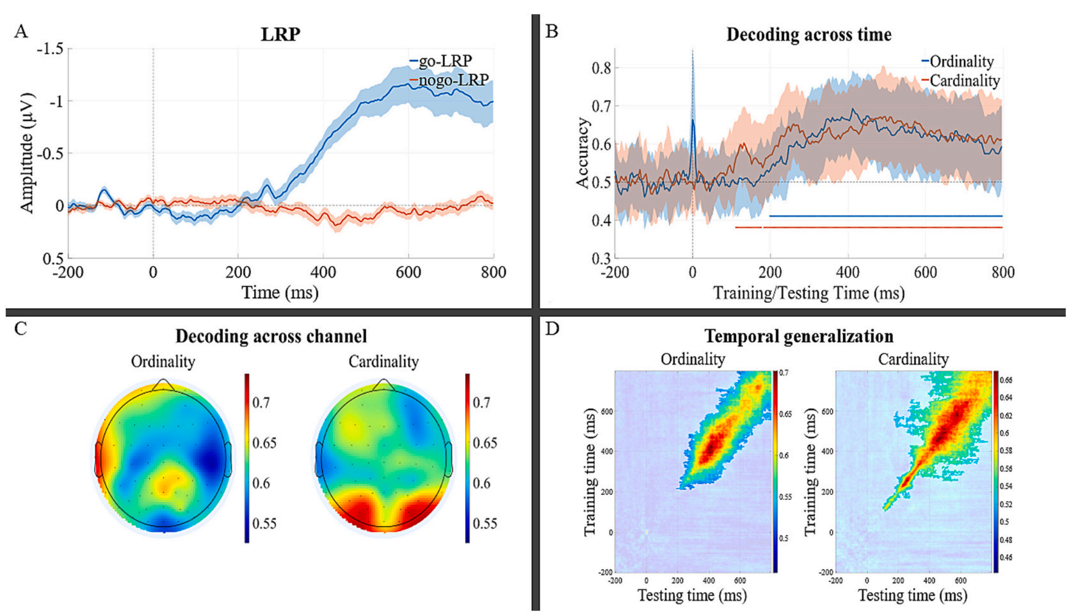

Fig. 6. Results of Univariate Analysis and MVPA of EEG in Experiment 2. Note: (A) Grand average LRPs for go and nogo trials. Shaded areas represent the standard error of the mean (SEM). Significant LRPs were observed in go trials, but not in nogo trials. B) Time-resolved decoding of numerical semantics information. Shaded areas indicate the SEM of decoding accuracy. The horizontal and colored solid lines at the bottom mark the time windows during which decoding accuracy for distinct semantics categories significantly differed from chance level (p < 0.05, corrected). C) Cross-electrode decoding of numerical semantics information. D) Temporal generalization results for numerical semantics decoding. Classifiers were trained at time points along the y-axis and tested at time points along the x-axis. Highlighted regions represent time windows where decoding accuracy was significantly higher than chance level (p < 0.05, corrected).

was observed only in Experiment 1, whereas the ordinal position effect emerged consistently across both experiments. EEG findings showed that traditional univariate ERP analysis did not detect significant activation of motor preparation related to numerical semantics. However, MVPA results revealed that participants processed cardinal semantics before ordinal semantics in the temporal dynamics. These findings suggest that the number order constructed in working memory primarily drives the SNAs and that there is a temporal dissociation between cardinal and ordinal semantics processing. This deepens our understanding of numerical cognition.

SNAs do not merely map quantity in space. Previous studies on the SNARC effect often attribute it to the varying spatial distributions of numerical symbols represented by numerical quantities (Dehaene, 2003; Feldman & Berger, 2022; Schwiedrzik et al., 2016). Animal studies have also identified similar sequential organization of quantity in spatial orientation. Since animals cannot learn symbols, they respond to intuitive non-symbolic quantities, supporting the view that spatially oriented representations based on numerical information are genetic and biological mechanisms (Giurfa et al., 2022; Rugani et al., 2015; Rugani et al., 2024). These viewpoints are derived from observations that animals exhibit a directional response preference to quantity dot arrays. For instance, bees move to the left when presented with fewer dots and to the right when presented with more dots. It is worth noting that this quantity-guided response preference is not direct evidence of SNAs. In many species, low spatial frequency stimuli are more readily observed on the left, while high spatial frequency stimuli are observed on the right (Frasnelli et al., 2012; Pitt et al., 2023). This phenomenon may originate from hemispheric biases in processing different spatial frequencies (left hemisphere biases toward high frequencies and right hemisphere toward low frequencies). Since numerosity often carries higher spatial frequency information, the spatial mapping of numerical magnitude might emerge as a byproduct of visual spatial frequency encoding (Felisatti et al., 2020). However, even when accounting for the potential confound between numerosity and spatial frequency cues, spatial frequency was not found to be the sole determinant of SNAs (Adriano et al., 2022). Moreover, SNAs are an implicit measure of individuals' asymmetric responses to numbers in different directions, rather than a selected response preference.

Experiment 1 behavioral results revealed significant ordinal position effect and SNARC effect, Experiment 2 only showed significant ordinal position effect. Notably, the patterns of ordinal position effect differed between the two experiments. Given the differences in task order where Experiment 2 required participants to perform parity judgments first, response hand assignments determined by whether the numbers belonged to the working memory sequence, these discrepancies may arise from task demands. The study by Ginsburg et al. (2014) also demonstrated that alterations in task demands can influence the retrieval of numerical information in working memory sequence. This different task demand effect influencing SNAs also extends to non-numerical sequences without cardinal meaning (e.g., letters). More importantly, temporarily constructed short-term sequences and over-learned sequences in long-term memory can coexist (Roth et al., 2025). Individuals typically default to extracting numerical sequences from long-term memory when processing numbers, and cultural and genetic influences result in numerical representations sharing the same directionality regarding order and quantity, with smaller numbers at the beginning of sequences and larger numbers at the end. Thus, when cardinal and ordinal semantics of numbers are confused, SNAs are wrongly attributed solely to the spatial representation of quantity. However, when individuals temporarily construct numerical sequences for tasks, they retrieve the sequences from working memory to complete parity judgment tasks. The construction of sequences is not rooted in the quantity of numbers. Thus, SNAs represent constructed numerical order relationships in space, not a direct mapping of quantity. Studies on the Tsimane indigenous people found that individuals who have not learned numerical sequences in social contexts do not arrange stimuli directionally according to quantity (Pitt et al., 2021). Therefore, quantity information is not the main factor determining SNAs.

The spatial representation of numbers primarily relies on the activation of ordinal semantics. Researchers in numerical cognition often use single digits as experimental stimuli to avoid confounding factors introduced by multiple digits. However, studies on double digits have revealed similar effects (Ganor-Stern et al., 2009; Huber et al., 2015; Weis et al., 2018; Zhou et al., 2008). In experiments, presumed larger numbers (whether 8 and 9 among single digits or 80 and 90 among double digits) are determined by comparison with a reference number, and their magnitude varies with the context. The brain interprets numerical magnitude relatively by constructing contextual ranking-space and reference points, integrating top-down task context with sensory inputs to situate isolated stimuli within a meaningfully framed mental space (Abrahamse & Van Dijck, 2023). Previous research often confuses cardinal and ordinal semantics, which makes it difficult to explain why similar effects are observed in stimuli containing only order information (such as letters and months) (Gevers et al., 2003, 2004). Behavioral findings reveal that SNAs emerge through the dynamic construction of numerical order in working memory, with the relative sequencing of number stimuli serving as a prerequisite for these effects. This flexibility is further evidenced by left-to-right readers reversing SNAs directionality through mental construction of a clock-like sequence during tasks (Bächtold et al., 1998; Mingolo et al., 2023). Critically, this aligns with evidence from non-numeric domains: Prpic et al. (2016) demonstrated that task demands dynamically recruit distinct cognitive mechanisms (Order-Related vs. Magnitude-Related Mechanisms), with ordinal processing dominating direct tasks and magnitude associations emerging in indirect tasks. Our findings refine this model by revealing that within numeric systems, task-contextualized ordinality competes with rather than fully suppresses magnitude-spatial mappings—a phenomenon evidenced by a significant SNARC effect was observed in Experiment 1. Thus, it can be inferred that ordinal semantics primarily drives the SNAs, also the influence of cardinal semantics should not be overlooked.

The most significant contribution of this study lies in clarifying the temporal dynamics of cardinal and ordinal semantics processing of numbers. Previous research could only infer implicit numerical processing during parity judgment tasks through experimental paradigms and behavioral results. However, this viewpoint lacks support from neuroscience evidence. This study used ERP technique to separate the temporal processing of cardinal and ordinal semantics. MVPA revealed that cardinal semantics have a temporal processing advantage over ordinal semantics. In two experiments, the decoding accuracy of cardinal semantics was significantly higher than the chance at 112 ms after stimulus presentation. This indicates that individuals can process quantity information within a very short time after stimulus presentation, and the processing temporal dynamics of cardinal semantics is not affected by task requirements. Quantity processing of symbolic numbers is an automatic process that occurs early in cognitive activities and is usually not directly influenced by experimental tasks (Imbo et al., 2012).

The mechanisms underlying SNAs have been debated extensively. Casasanto and Pitt (2019) proposed that ordinal semantics provides a comprehensive explanation for these effects, whereas cardinal semantics accounts for only a subset of evidence. This view has been challenged by Prpic et al. (2021), who argue that cardinal semantics still play a role, particularly in contexts where stimuli lack overlearned ordinal semantics or when task demands explicitly engage magnitude processing. However, subsequent discussions emphasize that magnitude influences these effects primarily by establishing ordinal relationships among stimuli (Pitt & Casasanto, 2022). Together, these studies suggest that while ordinality appears to dominate in most paradigms, the interaction between stimulus characteristics and task requirements may modulate the relative contributions of cardinal and ordinal semantics. MVPA of ordinal semantics showed that decoding accuracy was significantly higher than the chance at 180 ms after stimulus presentation in Experiment 1 and at 200 ms in Experiment 2. This suggests that individuals can obtain order information of stimuli as early as 180 ms after stimulus presentation. These temporal differences may arise because when tasks require individuals to first judge the order of stimuli, task priority prompts the brain to process ordinal semantics faster. Conversely, when tasks require individuals to first judge the parity of stimuli, the processing of ordinal semantics may need to wait until the processing of cardinal semantics is completed. This was further confirmed by univariate ERP analysis. Specifically, the onset latency of go-LRP in Experiment 2 was 80 ms longer than in Experiment 1, indicating that the processing of ordinal semantics affected motor preparation for numerical responses. Research on the SNARC effect has shown that this effect appears in the late response stage and lacks substantial evidence in the early processing stage (Gevers et al., 2006; Keus et al., 2005). These studies suggest that in the early stage of numerical cognition, during the process of quantity recognition, no significant interaction with spatial position was found. Although the processing time of ordinal semantics is later than that of cardinal semantics, its impact on numerical response preparation is closely related to order recognition.

Earlier research has demonstrated that the intensity of SNAs is positively correlated with extended RTs (Gevers et al., 2006). This phenomenon can be explained by the enhanced ordinal semantics processing of numbers, resulting in more robust connections and, consequently, necessitating additional time. The contributions of the two types of semantics to SNAs are distinct. Thus, the representation of numbers in space and the asymmetry of response positions essentially reflect the acquisition of sequential relationships between numbers. Our MVPA results revealed critical evidence of numerical processing at 112 milliseconds post-stimulus onset, with this temporal marker remaining consistent across task demands, indicating automatic activation of numerical magnitude. More importantly, the intrinsic association between numbers and space emerges through linking the temporal dynamics of ordinal semantics processing with response onset latency, thereby constructing a spatial scaffold for numerical sequential relationships. Our findings further demonstrate that the presence of sequential position effects does not negate SNARC effects, as both can coexist under specific task requirements. MVPA results showed temporal overlap in decoding both semantic dimensions, suggesting potential parallel processing pathways for numerical semantics in the brain. Consequently, cardinal and ordinal semantics are processed in parallel neural subspaces with temporal overlap, suggesting they compete rather than integrate during spatial-numerical association formation.

Decoding results based on scalp electrodes reveal differences in semantics processing across tasks. For example, the electrodes contributing most to the decoding of cardinal semantics in Experiment 1 are mainly concentrated in the left occipital lobe, while in Experiment 2, they are concentrated in both occipital lobes. The contribution of ordinal semantics to scalp electrodes across tasks shows greater variability. It should be clarified that electrode activity in decoding does not directly equate to brain region activation. Due to variability in cortical folding among individuals and significant differences in the amplitude of rhythmic waves of different frequencies across electrodes, decoding results based on electrodes can only indicate differences in processing patterns between the two semantics on the scalp, but cannot directly point to specific cortical regions (Bazanova & Vernon, 2014; Borrell, 2018; Haufe et al., 2014).

The semantics processing modes of numbers are dynamic, with ordinal and cardinal semantics exhibiting differences in decoding patterns under various experimental task contexts. Results from temporal generalization indicate that when tasks require memory sequence detection first (Experiment 1), ordinal semantics demonstrate significant temporal generalization characteristics after 200 ms, enabling decoding at any subsequent time point. In contrast, cardinal semantics are difficult to transfer across time. In tasks require parity judgments first (Experiment 2), ordinal semantics still possess a certain degree of temporal generalization performance, while cardinal semantics exhibit significant temporal generalization ability after 300 ms. Ordinal semantics perform better in temporal generalization, attributed to the reliance of ordinal information on working memory for retrieval in this study. Previous research has confirmed that the various stages of working memory are not independent of each other. Neural activity patterns detected in early sensation can be traced back throughout the entire sensory processing period, achieving reliable decoding within relatively distant time frames (Turoman et al., 2024). Experiment 2 also observed good generalization performance of cardinal semantics in the temporal dimension. This may be because the diagonal decoding of cardinal semantics before 300 ms reflects automatic processing of quantities by individuals, while the decoding similarity observed at various time points after 300 ms reflects decoding motor actions. Firstly, significant differences in neural activity patterns between stationary and moving states can persist for a period of time after movement (Yu et al., 2023). Secondly, the univariate ERP results in Experiment 2 verify that the onset time of motor preparation, as indicated by the go-LRP, overlaps significantly with the broadened time range of cardinal semantics temporal generalization. Finally, the results of Experiment 1 confirm that in tasks where memory sequence detection determine keystroke, the decoding of ordinal semantics exhibits stable characteristics at subsequent time points. In summary, ordinal and cardinal semantics are not independent cognitive activities for numbers. They are influenced by various factors such as task instructions and response modes, which jointly affect the encoding and stability of numerical representations in space by individuals (Gevers et al., 2010; Pinto et al., 2021). Essentially, these factors influence the processing modes of the two semantics.

5. Conclusion

This study employs a dual choice go/nogo paradigm combined with ERP technique and MVPA to explore the semantics processing mechanisms within SNAs, focusing on the dynamic characteristics of cardinal and ordinal semantics over time. The comprehensive analysis of behavioral results and ERP data unveils several pivotal findings that enrich our understanding of numerical cognition and offer new perspectives for future research. This study demonstrates that the acquisition of ordinal relationships between numbers may constitute a central mechanism underlying SNAs. In contrast to traditional theories emphasizing the dominance of magnitude representations, the research highlights the primacy of ordinal processing in SNAs. This finding not only identifies the processing of numerical ordinal semantics as critical to SNAs but also affirms the role of spatially anchored magnitude representations. Moreover, this study successfully dissociates the processing temporal dynamics of cardinal and ordinal semantics. Cardinal semantics are processed within a very short period after stimulus presentation, whereas the processing of ordinal semantics occurs relatively later. Combined with the results of LRP, this finding supports the notion that ordinal semantics primarily drives the SNAs and explains why SNAs typically emerge during later response stages. Furthermore, the study reveals that ordinal semantics exhibit superior temporal generalization, highlighting the key role of working memory in ordinal information retrieval. These findings pave new directions and provide insights for future research, contributing to a deeper understanding of numerical cognition.

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CRediT authorship contribution statement

Zhiwei Zhang: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Formal analysis, Data curation, Conceptualization. **Qingqing Chen:** Formal analysis, Data curation. **Chengtao Wang:** Formal analysis, Data curation. **Dongxue Zhao:** Writing – original draft. **Yun Pan:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors declare no generative AI tool was used for the preparation of the present manuscript.

Declaration of competing interest

The authors declare that no competing interests exist.

Data availability

I have included a link in the manuscript to access the data.

Appendix A. Appendix

In accordance with Reviewer 2's suggestion, we conducted an analysis of the MARC (Linguistic Markedness of Response Codes) effect (Nuerk et al., 2004). The MARC effect, a recognized phenomenon in numerical cognition, refers to the compatibility effect between a number's parity (odd/even) and response codes (e.g., left or right hand) during parity judgment tasks. Specifically, participants exhibit faster RTs when responding to odd numbers with their left hand and to even numbers with their right hand, whereas the reversed mapping (odd-right/even-left) results in slower responses. Given that our experimental design incorporated dynamically constructed working memory sequences, we additionally analyzed the interaction between numerical parity and the temporal ordering of working memory representations.

Results of Experiment 1: To investigate compatibility between numerical parity and the ordinal position of working memory sequences, RTs were analyzed using a repeated measures ANOVA of ordinal position × parity was conducted. The analysis showed a significant main effect of ordinal position, $F(1, 30) = 26.94$, $p < 0.001$, $\eta_p^2 = 0.47$. Responses to the initial numbers in the sequence ($M = 771.78$ ms, $SE = 15.06$ ms) were significantly faster than responses to the later numbers ($M = 795.79$ ms, $SE = 16.02$ ms). The remaining main effect and interaction were not significant, with p-values greater than 0.318, indicating no significant compatibility was observed between numerical parity and the ordinal position of working memory sequences.

To investigate compatibility between numerical parity and response hand, RTs were analyzed using a repeated measures ANOVA of response hand × parity was conducted. The main effects and interaction were not significant, with p-values greater than 0.058, indicating no MARC effect.

Results of Experiment 2: To investigate compatibility between numerical parity and the ordinal position of working memory sequences, RTs were analyzed using a repeated measures ANOVA of ordinal position × parity was conducted. The analysis showed no significant main effect of ordinal position, $p = 0.160$. The analysis showed a significant main effect of parity, $F(1, 29) = 8.53$, $p = 0.007$, $\eta_p^2 = 0.23$. Responses to the odd numbers ($M = 715.13$ ms, $SE = 18.79$ ms) were significantly slower than responses to the even numbers ($M = 687.30$ ms, $SE = 17.47$ ms). There was a significant interaction between ordinal position and parity, $F(1, 29) = 22.44$, $p < 0.001$, $\eta_p^2 = 0.44$. Post hoc analysis revealed no significant difference in RTs between parity for initial numbers, odd numbers ($M = 697.98$ ms, $SE = 17.53$ ms) slower than the even numbers ($M = 694.58$ ms, $SE = 16.79$ ms), $p = 0.745$. The effect was exclusively driven by later numbers, with significantly faster even numbers of responses ($M = 680.03$ ms, $SE = 19.11$ ms) compared to odd numbers ($M = 732.29$ ms, $SE = 20.86$ ms), $p < 0.001$. Due to our task design, the congruence effect observed here between ordinal position and numerical parity is operationally equivalent to the ordinal position effect identified in Experiment 2.

To investigate compatibility between numerical parity and response hand, RTs were analyzed using a repeated measures ANOVA of response hand × parity was conducted. The analysis showed no significant main effect of response hand, $p = 0.389$. The analysis showed a significant main effect of parity, $F(1, 29) = 9.34$, $p = 0.005$, $\eta_p^2 = 0.24$. Responses to the odd numbers ($M = 755.32$ ms, $SE = 17.27$ ms) were significantly slower than responses to the even numbers ($M = 731.98$ ms, $SE = 17.39$ ms). There was a significant interaction between response hand and parity, $F(1, 29) = 134.56$, $p < 0.001$, $\eta_p^2 = 0.82$. Post hoc analysis revealed significant difference in RTs between parity for left hand, odd numbers ($M = 714.95$ ms, $SE = 18.73$ ms) faster than the even numbers ($M = 776.79$ ms, $SE = 18.52$ ms), $p < 0.001$. For right hand, with significantly faster even numbers of responses ($M = 687.18$ ms, $SE = 17.45$ ms) compared to odd numbers ($M = 795.70$ ms, $SE = 16.80$ ms), $p < 0.001$, indicating the presence of MARC effect.

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